FIN2020 HW7 文杰 123090612

Q1

a. Compute q^1 and q^2

Asset 1 has price 2 and payoff vector $(2, 4)$. Asset 2 has price 1.25 and payoff vector $(1, 3)$. Therefore, by no-arbitrage pricing,

$$2 = 2q^1 + 4q^2$$

and

$$1.25 = q^1 + 3q^2.$$

Solve the system:

$$q^1 = 0.50, \qquad q^2 = 0.25.$$

b. Price of the bond

The bond pays $(1, 1)$ across the two states. Using the state prices,

$$P_b^0 = q^1 Z_b^1 + q^2 Z_b^2.$$

Thus,

$$P_b^0 = 0.50(1) + 0.25(1) = 0.75.$$

c. Price of the stock

The stock pays no dividends, but it can be sold for 2 in state 1 and 3 in state 2. Thus its payoff vector is $(2, 3)$.

Using the same state-price formula,

$$P_s^0 = q^1 P_s^1 + q^2 P_s^2 = 0.50(2) + 0.25(3) = 1.75$$

Q2

Part 1: Is this market complete?

Financial markets are considered complete if, for each possible future state, there exists a market for a contingent claim. Mathematically, this means the number of linearly independent securities (M) must equal the number of possible states (N).

In this scenario, there are 3 states ($N = 3$) and 3 securities ($M = 3$).

$$P = \begin{bmatrix} 3 & 7 & 8 \\ 1 & 2 & 9 \\ 7 & 16 & 25 \end{bmatrix}$$

$$\det(P) = 3(2 \times 25 - 9 \times 16) - 7(1 \times 25 - 9 \times 7) + 8(1 \times 16 - 2 \times 7) = 0$$

Since the determinant is exactly 0, the columns are linearly dependent. Specifically, the payoffs of Security C can be perfectly replicated by buying 47 units of Security A and shorting 19 units of Security B ($47 \times C_A - 19 \times C_B = C_C$).

Because the rank of the matrix is 2, there are only 2 linearly independent assets for 3 states. Therefore, **this market is not complete.**

Part 2: Introduce Call Options to Make the Market Complete

By Proposition 3, a necessary and sufficient condition to complete a market is the existence of an asset (or portfolio) whose payoff pattern distinguishes among all future states, allowing options to be written on it.

Looking at **Security A**, its payoffs across the three states are 3, 1, and 7. Because these values are all distinct, Security A successfully distinguishes among all future states.

Introduce a call option on Security A with a strike price of $K = 3$:

- **State 1:** $\max(3 - 3, 0) = 0$
- **State 2:** $\max(1 - 3, 0) = 0$
- **State 3:** $\max(7 - 3, 0) = 4$

The new asset (let's call it $Call_A$) has the payoff vector $\begin{bmatrix} 0 \\ 0 \\ 4 \end{bmatrix}$.

Now, substitute Security C (which was redundant) with our new Call Option and check the new payoff matrix:

$$P_{new} = \begin{bmatrix} 3 & 7 & 0 \\ 1 & 2 & 0 \\ 7 & 16 & 4 \end{bmatrix}$$

Check the determinant of this new matrix to verify linear independence:

$$\det(P_{new}) = 0 - 0 + 4(3 \times 2 - 7 \times 1) = -4$$

$$\det(P_{new}) = 4(6 - 7) = 4(-1) = -4$$

Because the determinant is non-zero, these three assets (Security A, Security B, and the Call Option on A with $K = 3$) are linearly independent. By introducing this call option, we now have 3 linearly independent securities for 3 states, making the market **complete**.